

Qasid Ahmed Aleem

 qasidaleem |
 qasidaleem@proton.me |  +92.336.283.7573

SUMMARY

A computational astrophysicist (MPhil) with a strong foundation in theoretical cosmology and 4.5 years of industry experience solving complex data problems as a Systems Engineer and Computer Vision Scientist. My professional work has equipped me with advanced skills in statistical inference, large-scale data analysis, and building robust computational pipelines. I am seeking a PhD position to return to my core research passion: investigating the fundamental challenges and open questions of the Lambda CDM model. I am particularly interested in applying modern data science techniques and rigorous statistical frameworks to problems in cosmological parameter estimation, large-scale structure, and alternative model testing

WORK EXPERIENCE

Data Scientist

August 2023 - Present

- Develop and implement end-to-end machine learning pipelines for predictive analysis, from data preprocessing and feature engineering to model training, validation, and deployment.
- Design and architect AI models to extract insights and forecast trends from large-scale, complex datasets, directly informing strategic decision-making.
- Engineer custom algorithms to solve specific business problems, applying advanced techniques in statistical analysis and machine learning.
- Utilize a stack including Python (Pandas, Scikit-learn, TensorFlow/PyTorch), SQL, and cloud platforms to manage and analyze big data.

System Engineer / Computer Vision Scientist

Jan 2021 - July 2023

- **Algorithm Development & Embedded Systems:** Designed, implemented, and optimized real-time control algorithms for embedded systems (Teensy 4.1 microcontroller) across multiple high-stakes engineering projects:
 - **Power Electronics:** Developed and deployed core algorithms including Sine Wave Pulse Width Modulation (PWM) and Space Vector Pulse Width Modulation for a 3kVA inverter and power supply unit, enhancing power efficiency and signal fidelity.
 - **Large-Scale Engine Control:** Architected the control logic and software for an Engine Control Unit (ECU) managing a **6.6 MW MTU cruise diesel engine**, creating a robust and precise control mechanism for a complex, high-energy system.
- **Computer Vision & 3D Tracking:** Led the algorithm development for a computer vision project utilizing OpenCV and YOLOv5. Engineered custom software for real-time object detection, tracking, and **distance calculation**, demonstrating expertise in translating 2D visual data into accurate 3D positional metrics.
- **Cross-Disciplinary Problem-Solving:** Applied a physics-first approach to solve complex problems in mechanics, thermodynamics, and electromagnetism, leveraging strong skills in C++, Python, and numerical modeling to bridge the gap between theoretical concepts and operational hardware.

Visiting Faculty

August 2021 - December 2021

- Co-taught **Observational Astrophysics** to 6th-semester Bachelor's students as a guest instructor alongside an Assistant Professor.
- Developed and conducted a practical laboratory component for the course, providing students with hands-on experience in astrophysical data analysis techniques.
- Held office hours and provided mentorship to students on academic and research-oriented projects.

EDUCATION

2020 – 2023 MPhil in Computational Astrophysics

Institute of Space Science and Technology, University of Karachi

GPA: 3.95/4.00

Thesis: Does the Cosmological Principle Hold True for Extragalactic Tests Performed at $z > 1.5$?

- Explored the cosmic quasar dipole, pinpointing discrepancies between the Λ CDM model and evidence of anisotropy.
- Analyzed extensive quasar datasets from the *SDSS* and *WISE* surveys.
- Implemented cutting-edge cosmological techniques to test the validity of the Cosmological Principle.

2019 – 2020 MSc in Space Science (Astrophysics)

Institute of Space Science and Technology, University of Karachi

90.33%

Thesis: Analyzing the Suitability of Variables in Mapping the Milky Way Galaxy

- Investigated distance calculation methodologies, with a focus on Cepheid variable stars.

2016 – 2018 BSc (Honours) in Space Science

Institute of Space Science and Technology, University of Karachi

82.41%

PRESENTATIONS

2021 **Aleem, Q. , Iqbal, T** *Analyzing the suitability of visible band classical cepheids to map the Milky Way.*

International Space Science Seminar, University of Karachi, Karachi, Pakistan.

TECHNICAL SKILLS

Programming Languages	Python (Pandas, NumPy, OpenCV), SQL, C/C++, MATLAB, GNU Octave
Machine Learning & AI	PyTorch, YOLO (v4+ Architectures), TensorFlow/Keras, Computer Vision, Predictive Modeling
Tools & Platforms	Linux, Git, Jupyter, Embedded Systems (Teensy)
Astrophysics Tools	Astropy, TOPCAT, Experience with SDSS, WISE, & other astronomical survey data

CONFERENCES AND WORKSHOPS

2023 **Attendee**, Bio-Symposium on Bio-engineering and CRISPR Technology
Usman Institute of Technology, Karachi, Pakistan.

2018 **Attendee**, International Space Science Seminar
Institute of Space Science and Technology, University of Karachi, Pakistan.

2017 **Attendee**, International Space Science Seminar
Institute of Space Science and Technology, University of Karachi, Pakistan.

2016 **Attendee**, International Space Science Seminar
Institute of Space Science and Technology, University of Karachi, Pakistan.

AWARDS AND HONORS

- 2020 **Gold Medal** for securing 1st Position in the Faculty of Science *University of Karachi*
2020 **Gold Medal** for securing 1st Position in the Department of Space Science *ISST*
2018 **Second Position** in Bachelor of Science (Honours) in Space Science *ISST*

COMMUNITY SERVICE

- 2023 **STEM Outreach Volunteer** *Karachi, Pakistan*
Invited to deliver a guest podcast and mentoring session for 8th-grade students, discussing astronomy, careers in science, and the role of women in STEM fields to inspire the next generation of scientists.
- Peer Mentor** *Institute of Space Science & Technology*
Provided academic guidance and support to junior university peers on curriculum expectations, research strategies, and career paths in space science.
Collaborated with fellow students to provide pro-bono assistance on coding and data analysis projects.
- 2014 **Volunteer Teacher** *The Citizens Foundation (TCF)*
Taught core subjects to underprivileged children, supporting their educational development and access to quality learning opportunities.